

■ Algo Trading System

Complete Logic Guide — Hinglish Mein

Instrument: NIFTY 50 • Strategy: 8-30 EMA • Mode: Paper Trading
Capital: ■1,50,000 • Risk per Trade: 1% • Max Trades/Day: 3

*Yeh document tumhare liye banaya gaya hai taaki tum poora system samajh sako —
kya ho raha hai, kyun ho raha hai, aur kaise ho raha hai.*

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Chapter 1: System Ka Overview

Yeh ek **automated paper trading system** hai jo NIFTY 50 index ke options mein trade karta hai. 'Paper trading' matlab — real paisa nahi lagta, lekin sab kuch exactly waise hi hota hai jaise real trading mein hota. Iska purpose hai 1 mahine ka genuine data collect karna — winners, losers, win rate — taaki real money lagate waqt confident ho sako ki strategy kaam karti hai.

System 4 main cheezein karta hai:

- **Monitor karta hai** — Nifty ka price har 5 minute mein
- **Signal dhundhta hai** — EMA 8 aur EMA 30 ke basis par
- **Filters lagate hai** — VIX, news, OI, candle patterns
- **Trade karta hai** — Option buy karta hai, SL/target track karta hai, EOD close karta hai

Files Ka Structure

File	Kaam Kya Hai
algo_trader.py	Main brain — poora loop yahan chalata hai
paper_trade.py	Trade engine — order place/close/track karta hai
option_chain.py	Kite API se real option price fetch karta hai
oi_data.py	OI levels (support/resistance) fetch karta hai
candle_patterns.py	Candle kaisi hai — dominance ya rejection
iv_filter.py	India VIX check karta hai
news_filter.py	Aaj koi bada event toh nahi?
config.py	Sab settings yahan hain — tum yahan hi changes karo
dashboard_server.py	Web dashboard chalata hai (localhost:4200)
report.py	Trade alerts print karta hai, daily JSON save karta hai

Chapter 2: Pre-Market Checks

Algo subah run karo toh 9:15 se pehle yeh 3 cheezein check karta hai. Agar koi bhi fail ho gayi, algo **poora din trade nahi karta**.

Check 1 — News Filter

news_filter.py check karta hai:

- Aaj market holiday toh nahi? (BSE/NSE calendar)
- RBI MPC meeting aaj hai? (Interest rate decision wale din volatility bahut zyada hoti hai)
- Koi NSE special event?

Agar haan — algo band ho jaata hai. Reason: intraday mein unexpected moves aate hain jo EMA strategy ko fool karte hain.

Check 2 — VIX Filter

India VIX = market ki 'fear gauge'. Jitna zyada VIX, utna zyada uncertainty.

VIX Range	Status	Position Size	Kya Matlab
Below 12	VERY LOW	100%	Market soya hua hai, trend nahi milega
12–20	NORMAL	100%	Sweet spot — EMA strategy best kaam karti hai
20–25	ELEVATED	50%	Thoda risk, half lots lo
25–30	HIGH	0% — Skip	Bahut volatile, mat khelo
30+	DANGEROUS	0% — Skip	Market mein aag lagi hai, bahar raho

Check 3 — OI Levels (Pre-Market)

Option chain se support aur resistance levels nikalte hain. Yeh sirf information ke liye hain — actual blocking trading hours mein hoti hai (Chapter 8 mein detail).

Opening Volatility Window — 9:15 to 9:45

Market khulne ke pehle 30 minute mein algo koi bhi trade nahi leta. Reason: opening mein bahut sare fake moves aate hain — institutions apni positions set karte hain, retail traders panic karte hain. Yeh moves often reverse ho jaate hain. 9:45 ke baad market settle ho jaata hai aur genuine trend shuru hota hai.

Chapter 3: EMA Strategy — Core Logic

EMA Kya Hota Hai?

EMA = Exponential Moving Average. Yeh price ka ek smoothed version hai jo recent prices ko zyada importance deta hai aur old prices ko kam. Simple Moving Average (SMA) se better hai kyunki yeh price changes par faster react karta hai.

Humara Setup: EMA 8 aur EMA 30

- **EMA 8** = Last 8 candles ka average — fast line, price ke karib rehta hai
- **EMA 30** = Last 30 candles ka average — slow line, trend dikhata hai

Condition	Direction	Matlab
EMA 8 > EMA 30	BUY (Bullish)	Short term average upar — uptrend hai
EMA 8 < EMA 30	SELL (Bearish)	Short term average neeche — downtrend hai

ADX — Trend Ki Strength Naapna

Sirf EMA cross kafi nahi. Market sideways bhi ho sakta hai jab EMA 8 aur EMA 30 ek doosre ke karib hote hain. Isliye ADX use karte hain.

ADX = Average Directional Index. Yeh 0-100 ke beech hota hai:

- ADX < 20 → Market sideways hai — algo trade nahi leta
- ADX 20-40 → Moderate trend — trade lete hain
- ADX 40+ → Strong trend — best conditions

Candle Patterns — Quality Check

Har signal mein candle ka shape bhi check hota hai:

Pattern	Condition	Matlab
Dominance Candle	Body $\geq$ 65% of candle range	Strong buyers/sellers — conviction hai
Rejection Candle	Wick $\geq$ 55% of candle range	Price ek level se wapas aaya — rejection hui
Bullish	Close > Open	Green candle — buyers jeet gaye
Bearish	Close < Open	Red candle — sellers jeet gaye

Candle Range = High - Low (candle ki poori height)

Body = |Close - Open| (sirf real body ki height)

Chapter 4: Signal Detection — Trade Kab Leta Hai

Do types ke entry signals hain. Dono mein ADX > 20 hona zaroori hai.

Entry Type 1 — Retest Entry (1:3 Risk:Reward)

Yeh tab hota hai jab price EMA 30 ke paas wapas aata hai aur ek achhi candle banaata hai.

Sochne ka tarika: EMA 30 ek 'road' ki tarah hai. Price usse door jaata hai, phir wapas aata hai (retest karta hai), aur phir phir se trend ki direction mein jaata hai. Yeh ek high-probability trade hai kyunki hum trend ke saath ja rahe hain.

Conditions for BUY Retest:

- EMA 8 > EMA 30 (uptrend)
- Price EMA 30 ke 0.4% ke andar hai
- Candle dominance ya rejection hai
- Candle bullish hai (green)
- ADX > 20

Conditions for SELL Retest:

- EMA 8 < EMA 30 (downtrend)
- Price EMA 30 ke 0.4% ke andar hai
- Candle dominance ya rejection hai
- Candle bearish hai (red)
- ADX > 20

Entry Type 2 — Continuation Entry (1:2 Risk:Reward)

Yeh tab hota hai jab EMA 8 aur EMA 30 bahut door ho jaate hain — matlab trend bahut strong hai — aur ek strong candle banaata hai. Hum trend ke momentum mein jump karte hain.

EMA Gap 'Stretched' kab hota hai?

Jab EMA 8 aur EMA 30 ke beech ka gap, current price ka 0.3% se zyada ho.

Formula: $|EMA8 - EMA30| / Price \geq 0.003$

Conditions for Continuation:

- EMAs stretched hain
- Dominance candle hai (strong body)
- Direction ke according bullish/bearish
- ADX > 20

SL aur Target Kaise Calculate Hote Hain

Parameter	BUY ke liye	SELL ke liye
Entry	Candle High + 0.5 pts buffer	Candle Low - 0.5 pts buffer

Stop Loss (SL)	Candle Low - 2 pts buffer	Candle High + 2 pts buffer
Target (Retest)	Entry + (Risk × 3)	Entry - (Risk × 3)
Target (Continuation)	Entry + (Risk × 2)	Entry - (Risk × 2)

Risk = |Entry - SL| (ek trade mein kitna point risk le rahe hain)

Chapter 5: Option Trading — Spot Se Option Mein Convert

Algo directly Nifty index mein trade nahi karta — woh **options khareedta hai**. Kyun? Kyunki options mein limited risk hoti hai (sirf premium lost ho sakta hai) aur leverage milta hai (thode paison mein bada position).

Option Type Kaunsa Lete Hain?

Signal Direction	Option Type	Kyun
BUY (Bullish)	CALL option (CE)	Call ka price badhta hai jab Nifty upar jaata hai
SELL (Bearish)	PUT option (PE)	Put ka price badhta hai jab Nifty neeche jaata hai

ATM Strike Kya Hai?

ATM = At The Money. Jis strike price par option ka premium sabse realistic hota hai (na bahut deep in-the-money, na bahut out-of-the-money). Hum hamesha ATM option lete hain.

NIFTY ATM Calculation:

```
Strike = Round(Spot / 50) × 50
Example: Nifty = 24,341 → ATM = Round(24341/50)×50 = 487×50 = 24,350 CE/PE
```

Delta — Spot Se Option Price Ka Connection

Delta batata hai ki Nifty 1 point move karne par option premium kitna move karega.

ATM option ka delta ≈ 0.5 (approximately)
Matlab: Nifty 10 points upar gaya → Call option premium ≈ 5 रुपये badhega

Spot SL/Target ko Option SL/Target mein convert karna:

```
Option SL = Option Entry - (Spot Risk × 0.5)
Option Target = Option Entry + (Spot Reward × 0.5)
```

Example:

Nifty spot entry = 24,400 | Spot SL = 24,390 | Spot Target = 24,430
Spot Risk = 10 pts | Spot Reward = 30 pts

```
Option Entry = ₹248 | Option SL = 248 - (10×0.5) = ₹243 | Option Target = 248 + (30×0.5) = ₹263
```

Expiry — Kaunse Week Ka Option?

NIFTY ke weekly options har Thursday ko expire hote hain. Expiry ke din zero-value contracts se bachne ke liye algo automatically agle hafte ka option select karta hai.

Lot Size — NIFTY ka 1 lot = 65 shares (Nov 2024 ke baad revised)

Chapter 6: Position Sizing — Kitne Lots Khareedne Hain

Yeh sabse important chapter hai. Galat position sizing se poori capital khatam ho sakti hai. Humara system do rules se protect karta hai:

Rule 1 — Risk-Based Sizing (1% per trade)

Ek trade mein maximum 1% capital risk ho sakti hai = ₹1,500 (on ₹1.5L capital).

Formula:

```
Risk Amount = Capital × 1% = ₹1,500
Risk Per Lot = (Option Entry - Option SL) × Lot Size
Lots = Risk Amount / Risk Per Lot
```

Example:

Option Entry=248, Option SL=238, Lot Size=65

Risk Per Lot = (248-238) × 65 = ₹650

```
Lots = 1500 / 650 = 2 lots (rounded down)
```

Rule 2 — Capital Cap (30% max per trade)

Kabhi kabhi spot SL bahut tight hoti hai (5-6 pts), jisse opt_risk_per_lot bahut kam hoti hai, aur lots count bahut zyada ho jaata hai. Isliye hard cap lagaya hai:

```
Max Lots by Capital = (Capital × 30%) / Lot Value
Final Lots = min(Risk-based lots, Capital-cap lots)
Example: ₹1.5L × 30% = ₹45,000 max | Lot Value = ₹16,120 → Max 2 lots
```

VIX Multiplier

Agar VIX elevated hai, lots automatically reduce hote hain:

VIX Level	Multiplier	Effect
Below 20	1.0 (100%)	Normal sizing
20–25	0.5 (50%)	Half lots
Above 25	0.0 (0%)	Trade skip

Chapter 7: Paper Trade Engine

paper_trade.py ek virtual broker ki tarah kaam karta hai. Koi real order Zerodha tak nahi jaata — sab simulation mein hota hai. Lekin prices 100% real hain — Kite API se live fetch hote hain.

Trade Ka Lifecycle

Step	Kya Hota Hai	Kahan Se Data
1. Entry	Trade create hota hai, premium store hota hai	Kite API — live LTP
2. Monitoring	Har candle par high/low check hota hai spot ke against	Kite API — live candles
3. SL Hit	Spot candle low $\leq$ Spot SL $\rightarrow$ exit trigger	Kite API — live option LTP at that moment
4. Target Hit	Spot candle high $\geq$ Spot Target $\rightarrow$ exit trigger	Kite API — live option LTP at that moment
5. EOD Close	15:20 par sab open trades close	Kite API — final live LTP
6. CSV Save	Trade record permanent CSV mein	logs/daily_report.csv

PnL Calculation

$$\text{PnL} = (\text{Exit Price} - \text{Entry Price}) \times \text{Qty}$$

$$\text{Total Cost} = \text{Entry Price} \times \text{Qty}$$

$$\text{Exit Value} = \text{Exit Price} \times \text{Qty}$$

Example: Entry=248, Exit=263, Qty=130 (2 lots $\times$ 65)

$$\text{PnL} = (263 - 248) \times 130 = 15 \times 130 = +\text{₹}1,950$$

Safety Guards

- MAX_TRADES_DAY = 3 $\rightarrow$ 3 se zyada trades nahi
- DAILY_LOSS_LIMIT = ₹4,500 $\rightarrow$ itna loss ho jaaye toh band
- Daily reset $\rightarrow$ naya din, naye counters (PnL, trade count, open trades clear)

Chapter 8: OI Filter — Support/Resistance

OI = Open Interest. Yeh batata hai ki kitne contracts abhi bhi open hain. Jis strike par sabse zyada Put OI hai → woh support hai (log wahan puts khareed kar neeche jana nahi chahte). Jis strike par sabse zyada Call OI hai → woh resistance hai.

OI Se Support/Resistance Kaise Milti Hai?

- Algo current spot ke aas paas ke ± 10 strikes scan karta hai
- Har strike ka PE OI aur CE OI Kite se fetch karta hai
- Sabse zyada PE OI wali strike = Support
- Sabse zyada CE OI wali strike = Resistance

Trade Block Kab Hota Hai?

Signal	Block Condition	Logic
BUY	Price $\geq$ Resistance - 50 pts	Resistance ke paas BUY risky hai — ceiling hai wahan
SELL	Price is right at Support wall (± 50 pts)	Support pe sell risky — floor tod bhi sakta hai
SELL	Price already below support by > 50 pts	Support toot chuka — freely trade karo

OI Refresh — Har 30 Minute Mein

OI static nahi hota — institutions positions change karte hain during the day. Isliye har 30 minute mein fresh OI levels fetch karta hai. Agar Kite unavailable ho toh OI filter skip ho jaata hai — trade block nahi hota.

Chapter 9: Sab Filters Ek Jagah

Ek trade tab tak nahi hota jab tak yeh saare gates clear na ho jaayein:

Filter	Source	Block Condition	Adjustment
Holiday	news_filter.py	Aaj market band hai	Poora din skip
RBI/Event	news_filter.py	High-impact event aaj hai	Poora din skip
VIX	iv_filter.py	$VIX \geq 25$	Skip day VIX 20-25 → half lots
Opening Window	news_filter.py	Time < 9:45 AM	Wait karo
ADX	algo_trader.py	$ADX < 20$	Sideways market — skip candle
No Trade After	algo_trader.py	Time $\geq 15:00$	No new entries
Candle Pattern	candle_patterns.py	Not dominance or rejection	Skip signal
OI Wall	oi_data.py	Price near max-OI level	Skip signal
Liquidity	algo_trader.py	Volume < 100 OR OI < 500	Option illiquid — skip
Max Trades	paper_trade.py	Trade count ≥ 3	No more trades today
Daily Loss	paper_trade.py	Day loss $\geq$ ₹4,500	Stop trading today

Chapter 10: Daily Flow — Subah Se Shaam Tak

Time	Kya Hota Hai
Before 9:15	python3 zerodha_login.py → fresh access token lo
Before 9:15	python3 algo_trader.py → algo shuru karo
Before 9:15	Pre-market checks: News → VIX → OI levels
9:15	Market open — algo loop shuru
9:15–9:45	Opening window — sirf monitor, koi trade nahi
9:45 onwards	Har 5 min: candle fetch → indicators → signal check → filters → trade?
Every 30 min	OI levels refresh (spot ke aas paas ke strikes)
Every 5 min	Open trades ka SL/Target check karo
Anytime	SL hit ya Target hit → live Kite LTP se exit, CSV mein save
15:00	Koi nayi entry nahi (NO_TRADE_AFTER)
15:20	Market close — sab open trades Kite live LTP pe close
15:20	Daily report print + JSON save in logs/
15:20	python3 dashboard_server.py → browser mein dekho

Chapter 11: Config File — Tumhare Haath Mein Kya Hai

config.py ek file hai jise tum directly edit kar sakte ho bina rest of the code ko samjhe. Yahan sab important settings hain:

Setting	Current Value	Kya Karta Hai	Suggestion
CAPITAL	■1,50,000	Starting capital	Real money mein same rakhna
RISK_PCT	0.01 (1%)	Risk per trade	1% safe hai beginners ke liye
MAX_TRADES_DAY	3	Max trades/day	3 enough hai — quality over quantity
DAILY_LOSS_LIMIT	■4,500	Day stop-loss	3 trades × ■1,500 = ■4,500
TIMEFRAME	5 minutes	Candle size	5 min best for intraday options
EMA_FAST	8	Fast EMA period	Standard — change nahi karna
EMA_SLOW	30	Slow EMA period	Standard — change nahi karna
ADX_THRESHOLD	20	Min trend strength	20 safe — 15 par zyada trades
OI_BUFFER	50 pts	OI wall buffer zone	50 best — 100 zyada block karta tha
NO_TRADE_AFTER	15:00	Last entry time	15:00 safe — expiry risk avoid

Suggestion: Paper trading complete hone ke baad in settings ko backtest results ke basis par fine-tune karna. Ek mahine baad tumhare paas real data hoga — tab decide karna ki ADX threshold badhana hai ya kam karna, risk 1% rakhna hai ya 1.5% karna.

Appendix: Sabhi Words Ka Matlab

Yahan har woh word explain kiya gaya hai jo document ya logs mein aata hai.

ADX (Average Directional Index)

Trend ki strength naapne wala indicator. 0-100 ke beech. 20 se upar ho toh trending market. Sideways market mein ADX neeche hota hai — tab algo trade nahi leta.

ATM (At The Money)

Woh option strike jo current spot price ke sabse karib hoti hai. Example: Nifty 24,341 par hai toh ATM strike 24,350 hogi.

Bid / Ask

Bid = sabse zyada price jo buyer dene ko ready hai. Ask = sabse kam price jo seller lene ko ready hai. Tum hamesha Ask par khareedoge aur Bid par bechoge.

Call Option (CE)

Ek contract jo tumhe right deta hai Nifty ko ek fixed price par khareedne ka. Jab Nifty upar jaata hai, Call ka premium badhta hai.

Capital

Trading ke liye available total paisa. Hamare case mein ₹1,50,000.

Continuation Entry

Jab EMA 8 aur EMA 30 bahut door ho jaate hain (stretched) aur ek strong candle bane, tab trend ke saath trade karte hain. 1:2 risk:reward.

Delta

Option ka sensitivity measure. ATM option ka delta ≈ 0.5 matlab Nifty 1 point move kare toh option premium 0.5 point move karega.

Dominance Candle

Ek candle jiska body (open-close gap) overall range (high-low) ka 65% ya zyada ho. Strong directional move dikhata hai.

EMA (Exponential Moving Average)

Price ka weighted average jisme recent prices ko zyada importance dete hain. EMA 8 = fast, EMA 30 = slow.

EOD (End of Day)

Trading day ka end. 15:20 IST par algo sab open positions close kar deta hai.

Expiry

Woh date jab option contract expire ho jaata hai. NIFTY weekly options har Thursday expire hote hain.

India VIX

NSE ka volatility index. Market mein kitna fear hai uska measure. 20 se neeche = normal, 25 se upar = dangerous.

Kite API

Zerodha ka programming interface. Isse real-time prices, candles, option data fetch karte hain.

Liquidity

Option mein kitna trading volume aur open interest hai. Kam liquidity wale options avoid karte hain kyunki bid-ask spread bahut wide hoti hai.

Lot

Options ek fixed quantity mein aate hain. NIFTY ka 1 lot = 65 shares.

Lot Value

Ek lot ki total cost. = $LTP \times Lot\ Size$. Example: $248 \times 65 = 16,120$.

LTP (Last Traded Price)

Woh price par sabse recent trade hua. Yahi hamare entry/exit price hote hain.

MPC (Monetary Policy Committee)

RBI ki committee jo interest rates decide karti hai. Iske meetings wale din market volatile hoti hai.

OI (Open Interest)

Abhi kitne option contracts open hain (na beca, na expire hua). High PE OI = support. High CE OI = resistance.

Paper Trading

Simulated trading — real paisa nahi lagta lekin sab calculations real prices se hoti hain. Strategy test karne ka best tarika.

PCR (Put-Call Ratio)

Total Put OI / Total Call OI. $PCR > 1$ = zyada puts = bullish sentiment (hedging). $PCR < 1$ = zyada calls = bearish sentiment.

Put Option (PE)

Ek contract jo tumhe right deta hai Nifty ko ek fixed price par bechne ka. Jab Nifty neeche jaata hai, Put ka premium badhta hai.

Rejection Candle

Ek candle jiska wick (shadow) overall range ka 55% ya zyada ho. Price ek level se reject hua — wapas aa gaya.

Resistance

Woh price level jahan sellers strong hote hain aur price upar jaane se ruk jaati hai. Max CE OI wali strike.

Retest Entry

Jab price EMA 30 ke paas wapas aata hai aur bounce karta hai. High probability setup. 1:3 risk:reward.

Risk:Reward (RR)

Risk kitna hai vs Reward kitna mil sakta hai. 1:3 matlab 1 risk par 3 ka potential profit.

SL (Stop Loss)

Woh price jahan par agar market jaaye, trade automatically close ho jaata hai aur loss book ho jaata hai.

Spot Price

Nifty index ka actual current price. Options is price se derive hote hain.

Strike Price

Woh fixed price jo option contract mein hota hai. Example: NIFTY 24400 CE mein 24400 = strike.

Support

Woh price level jahan buyers strong hote hain aur price neeche jaane se ruk jaati hai. Max PE OI wali strike.

Target

Woh price jahan par trade automatically close ho jaata hai aur profit book hota hai.

Trend

Price ki consistent direction — ya toh upar (uptrend/bullish) ya neeche (downtrend/bearish).

VIX

Volatility Index. Dekho India VIX.

Weekly Expiry

NIFTY options har hafte Thursday ko expire hote hain. BANKNIFTY bhi weekly.